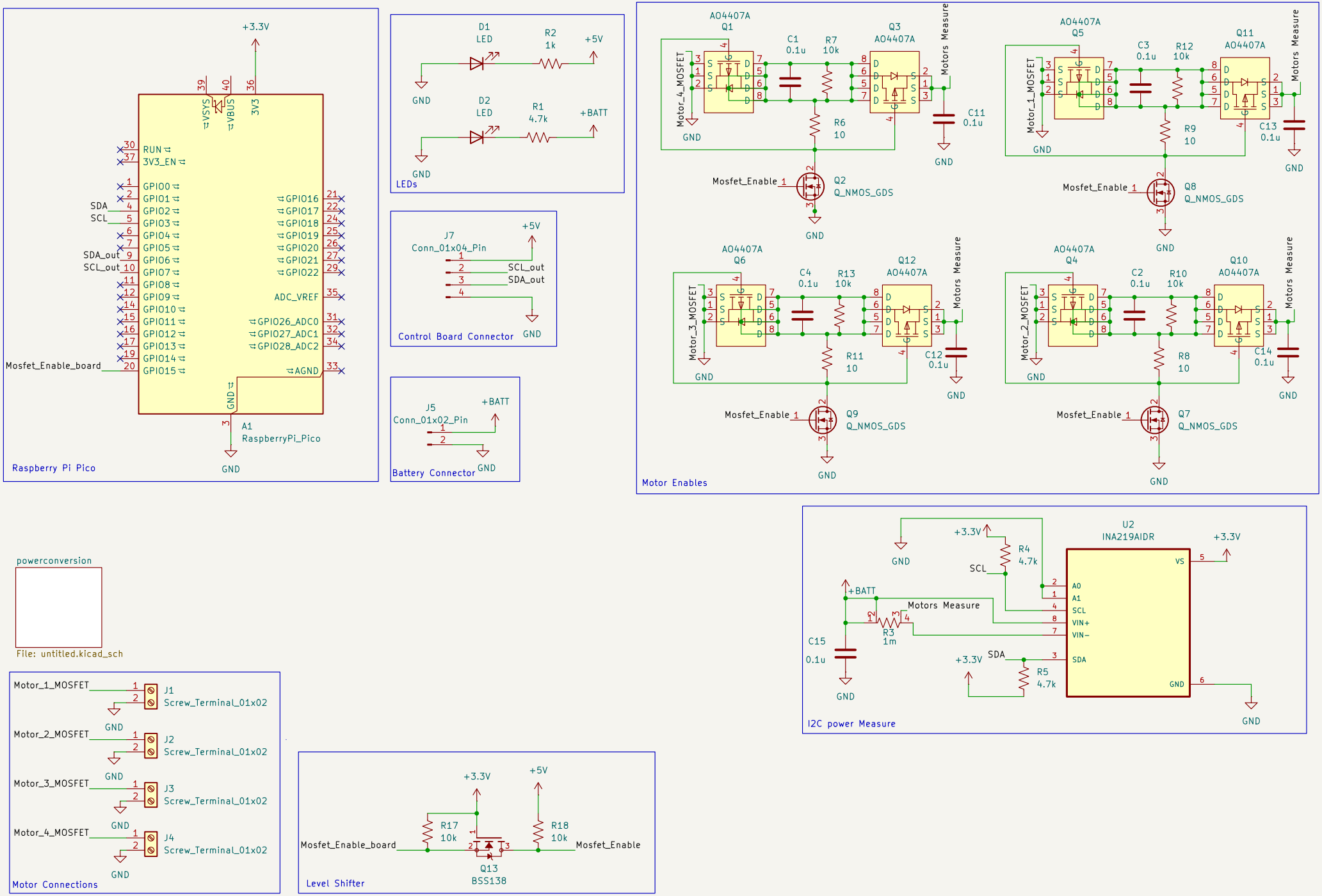
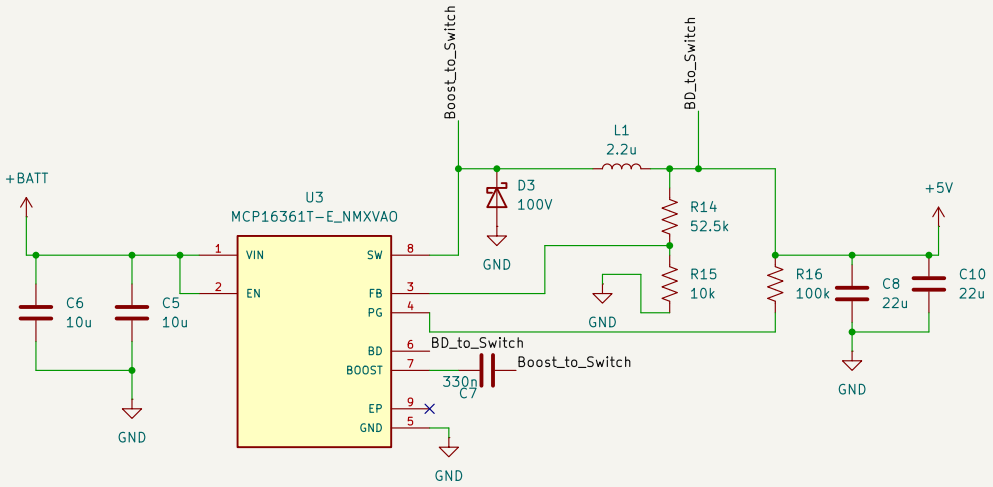
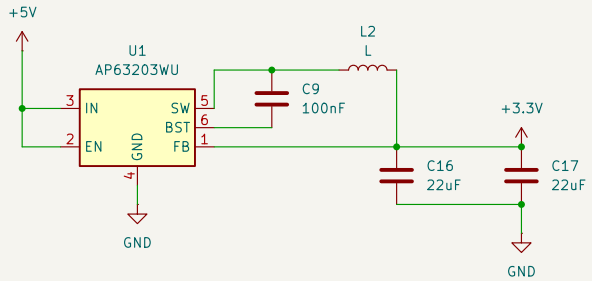

Raspberry Pi Pico
A1
RaspberryPi_Pico
+3.3V
GND
RUN
3V3_EN
GPIO0
GPIO1
GPIO2
GPIO3
GPIO4
GPIO5
GPIO6
GPIO7
GPIO8
GPIO9
GPIO10
GPIO11
GPIO12
GPIO13
GPIO14
GPIO15
GPIO16
GPIO17
GPIO18
GPIO19
GPIO20
GPIO21
GPIO22
ADC_VREF
GPIO26_ADC0
GPIO27_ADC1
GPIO28_ADC2
AGND
VSYS
VBUS
3V3
SDA
SCL
SDA_out
SCL_out
Mosfet_Enable_board
LEDs
D1
LED
R2
1k
+5V
D2
R1
4.7k
+BATT
Control Board Connector
J7
Conn_01x04_Pin
SCL_out
SDA_out
Battery Connector
J5
Conn_01x02_Pin
Motor Enables
AO4407A
Q1
Q2
Q3
Q4
Q5
Q6
Q7
Q8
Q9
Q10
Q11
Q12
Q_NMOS_GDS
Motor_1_MOSFET
Motor_2_MOSFET
Motor_3_MOSFET
Motor_4_MOSFET
Motors Measure
Mosfet_Enable
C1
C2
C3
C4
0.1u
C10
C11
C12
C13
C14
R6
R7
R8
R9
R10
R11
R12
R13
10
10k
powerconversion
File: untitled.kicad_sch
Motor Connections
J1
J2
J3
J4
Screw_Terminal_01x02
Level Shifter
R17
R18
Q13
BSS138
I2C power Measure
U2
INA219AIDR
R3
1m
R4
R5
4.7k
C15
VS
A0
A1
SCL
VIN+
VIN−
SDA


+BATT
U3
MCP16361T-E_NMXVAO
VIN
EN
SW
FB
PG
BD
BOOST
EP
GND
1
2
8
3
4
6
7
9
5
C6
10u
C5
10u
GND
Boost_to_Switch
BD_to_Switch
L1
2.2u
D3
100V
R14
52.5k
R15
10k
R16
100k
C8
22u
C10
22u
+5V
330n
C7
BD_to_Switch
Boost_to_Switch
GND
GND
GND


+5V
U1
AP63203WU
IN
EN
GND
SW
BST
FB
3
2
4
5
6
1
C9
100nF
L2
L
+3.3V
C16
22uF
C17
22uF
GND
GND